

Introduction to Tries

1 What Is a Trie?

Trie

A **trie** is a tree-based data structure used to store a collection of strings. Each edge represents a character, and each root-to-node path represents a prefix.

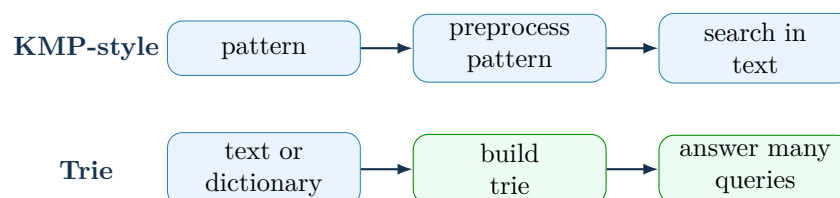
The name is associated with **retrieval**: a trie makes it efficient to retrieve words, prefixes, and dictionary entries.

A trie is useful when:

- the dictionary or text collection is fixed;
- many search queries must be answered;
- prefix queries are important;
- many words share common prefixes.

2 Trie vs Pattern-Preprocessing

String matching algorithms such as KMP preprocess a pattern, then search inside a text. A trie takes the complementary viewpoint: preprocess the stored collection, then answer many query strings.



When a Trie Fits

Use a trie when the stored collection changes slowly but prefix or word queries are frequent.

3 Main Idea: Prefix Sharing

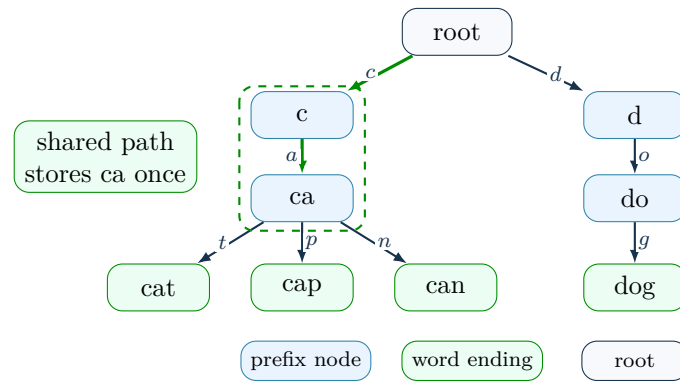
Suppose the stored words are

cat, cap, can, dog.

The words cat, cap, and can all start with the same prefix:

ca.

In a trie, that prefix is stored once, not three times.



4 Path Meaning

Every node represents the string spelled by the path from the root to that node.

Path	String represented
$\text{root} \rightarrow c$	c
$\text{root} \rightarrow c \rightarrow a$	ca
$\text{root} \rightarrow c \rightarrow a \rightarrow t$	cat
$\text{root} \rightarrow d \rightarrow o \rightarrow g$	dog

5 Complete-Word Matching

Complete-Word Search

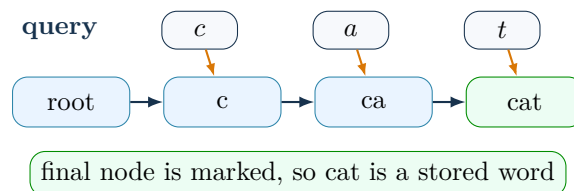
To check whether a word is stored, follow one edge per character. The word is present only if the whole path exists and the final node is marked as a word-ending node.

For the word

cat,

the search path is

$\text{root} \rightarrow c \rightarrow a \rightarrow t$.



6 Prefix Matching

Prefix Search

To check whether a prefix exists, follow one edge per prefix character. If the path exists, then at least one stored word begins with that prefix.

For prefix

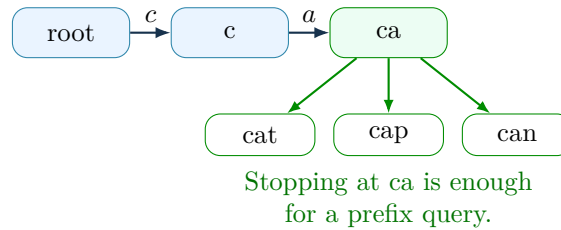
ca,

the search path is

root $\rightarrow$ c $\rightarrow$ a.

Since the path exists, the dictionary contains at least one word beginning with ca:

cat, cap, can.



Prefix Result

Path for ca exists $\Rightarrow$ there is at least one stored word with prefix ca.

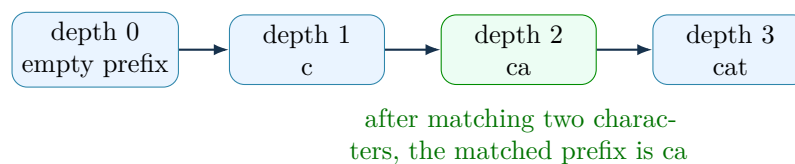
7 Depth Interpretation

Depth Rule

If a search reaches depth k , then exactly the first k characters of the query string have been matched.

depth $k \iff k$ matched characters.

For example, reaching the node for prefix ca means the search reached depth 2, so the first two characters were matched.



8 Basic Operations

Operation	Idea	Cost
Insert word w	Create missing edges while scanning characters of w .	$\mathcal{O}(w)$
Search word w	Follow existing edges for all characters and check word-ending mark.	$\mathcal{O}(w)$
Search prefix p	Follow existing edges for all prefix characters.	$\mathcal{O}(p)$

The cost depends on the length of the query string, not directly on the number of stored words.